

OPEN SOURCE HARDWARE

A Golden Opportunity For Innovators And Small Manufacturers

Would it make sense to extend the logic of free/open source software to the hardware field? Would it be of any help if the reference designs of chipsets and microcontrollers were available under free licences for all to use? Is India capable of using these to build exciting commercial products?

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Once upon a time, not too long ago, Massimo Banzi, a teacher at a high-tech design school in Italy felt constrained by the fact that neither he nor his students could find an affordable yet powerful microcontroller to build their robots. In 2005, Banzi met David Cuartielles, a Spanish microchip engineer, and shared his team's problem with him. The two decided to work together to design their own board, seeking the help of a student called David Mellis to write the programming language for it. In three days, the board was ready. The students loved it. It was so easy to use that even those who did not know much programming could use it to do some cool stuff.

Banzi, Cuartielles and Mellis 'open sourced' the reference design, posted the schematics online and made 200 copies of the board, hoping somebody would buy it. The school they worked for bought 50, and they were scared about whether the remaining would sell. After all, they had spent 3000 Euros on it! Not only did the remaining 150 boards sell but within a few months orders started pouring in for hundreds more. The board—Arduino—grew in popularity and is a popular buzzword in open source hardware today.

A beautiful piece of engineering. Extremely useful. Huge market. Com-



Neo FreeRunner open-source smartphone from OpenMoko

mercially successful. Why then did the team decide to 'open source' the reference design of the board they sold instead of protecting it as a proprietary design? Would they not have earned much more if they ensured that nobody else could reuse their design to make and sell similar boards?

Well, being an academic not too interested in money, Cuartielles might have favoured the open sourcing to

inspire creativity in students and encourage more people to use the board for experiments. However, Banzi, as described in a *Wired* magazine story, was more of a canny businessman. His decision to open source the reference designs and the resultant success of their product, is a lesson to all in the computer and electronics industries. Banzi rightly believed that open sourcing Arduino would result in greater adoption, publicity and the increased participation of innumerable intelligent engineers in improving the design.

Banzi's hunch proved to be right. Within two years, 30,000 units of the Arduino board were sold. The board began to be deployed for varied purposes, ranging from fine-tuning car engines and monitoring houseplants, to building laptops and unmanned model planes. The wide usage was not merely because the Arduino board was cheaper than its counterparts but because it was open and users could hack it and harness it to suit their specific requirements. Users and others offered suggestions to improve the design and the programming language. Every version of the board sold more and was better than the previous ones. Since the Creative Commons licence they chose required anybody who used the design to credit the Arduino team in the product, Banzi and the others also came to be recognised as experts in the field, and became sought after consultants. In short, the team and their small company